SIMATSENGINEERING(SAVEETHASCHOOLOFENGINEERING)

DepartmentofComputerScienceandEngineering

ASSESSMENTTOOL1–INTERACTIVETREESIMULATOR

CourseCode:	CSA0305	**CourseTitle:**	DataStructures
COAssessed:	CO3–Precision(BL3,BL5)	**Assessment:**	AssessmentTool1–Interactive Tree Simulator
Weightage:	30%of CIA	**TotalMarks:**	100
CO3Statement:	*SolveproblemsforoptimizeddatastorageandretrievalusingBinarySearchTrees,AVLTrees,B-Trees, Red-Black Trees, and Splay Trees.*		
StudentName:	H SARAVANA	**Reg. No.:**	192511397
Section/Batch:	CSE	**Date:**	29/07/2026

CodeofConduct

I H SARAVANA (192511397)____(Name/RegNo) certifythatthissubmissionismyoriginalworkandthatIhaveadheredtotheguidelinesspecifiedforthis assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

SignatureoftheStudent:H Saravana

Instructions

- Thisassessmenttoolcontains5InteractiveTreeSimulatorquestions,Total:100Marks.
- Studentscanuseanyonline/offlineInteractiveTreeSimulator.Demonstrateinsertion,deletion, searching, and traversal operations wherever applicable.
- Whenastudentanswerusingsimulator,inevaluationtheanswershouldhavecapturedscreenshots after every major operation.
- SubmitthereportinPDFformatwithproperlylabelledscreenshotsasproofofcompletion.

Section/Topic	Focus	QNos	Marks
BinarySearchTree	Properties,binarytreeandbinarysearchtree,insertion and deletion	1	20
Tree Traversals	Traversaltypes,Inorder,Post orderandPreorder	2	20
AVLTree	Balancingfactor, Rotation,Typesofrotation,examples	3	20
BTree,2-3tree,2-3-4tree	Properties,structuresandOperations	4	20
RedBlackTree	Properties,ExampleandOperations	5	20
		GRANDTOTAL:	**100Marks**

AnnexureA–InteractiveTreeSimulator

1. ConstructaBinarySearchTree(BST)byinsertingthefollowingkeys:50,30,70,20,40,60,80

   Perform:

   a. InorderTraversal
   b. Searchfor60
   c. Delete30
   d. DisplaythefinalBST.

2. CompleteTreeTraversalsbyInserting50,30,70,20,40,60,80.Findallthetraversals.

   a. Inorder
   b. Preorder
   c. Postorder

3. Insert70,50,90,40,60,80,100,55intoanAVLTree.Delete40and90.Analyzehowthetree rebalances after each deletion.
4. ConstructaB-Treeoforder3byinserting45,25,65,15,35,55,75
5. Insert100,80,120,60,90,110,130intoaRed-BlackTree.Analyzetheirheights.

Rubricsfor Calculation

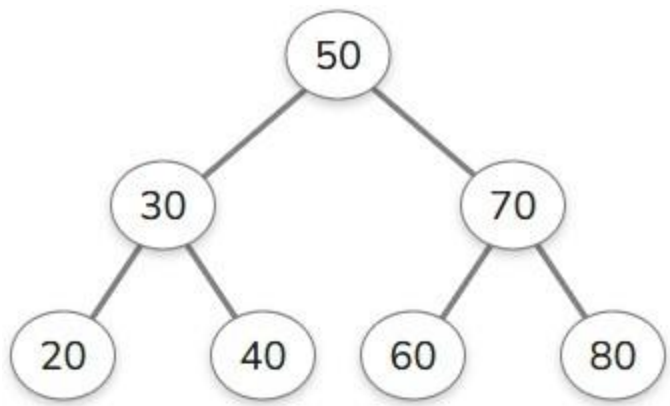
Criteria	Weightage	Level4–Exemplary	Level3–Proficient	Level 2 – Developing	Level 1 – Beginning
CorrectTree Construction	20	Tree is constructedcorrectly with alloperationsperformedaccurately	Minorerrorsinconstruction	Someoperationsincorrect	Treeconstructionmostlyincorrect
UnderstandingofTreeOperations	20	Clearlyexplainsinsertion,deletion, search andbalancingoperations	Explainsmostoperationscorrectly	Limitedunderstanding	Unabletoexplainoperations
AnalysisofTree Behaviour	30	Correctlyanalysesbalancing, rotations, nodesplit/merge,splaying	Goodanalysiswith minoromissions	Partialanalysis	Poororincorrectanalysis
Presentation	20	Well-organizedreportwithaccurate observations andconclusion	Goodreportwith minorissues	Basic report withlimitedobservations	Incompletereport
AccuracyofResponses	10	90–100%correctanswers	75–89%correctanswers	50–74%correctanswers	Below50%correctanswers

FacultyIn-charge	CourseCoordinator	ProgramDirector
Signature: ______________	Signature: ______________	Signature: ______________
Date: _________________	Date: _________________	Date: _________________

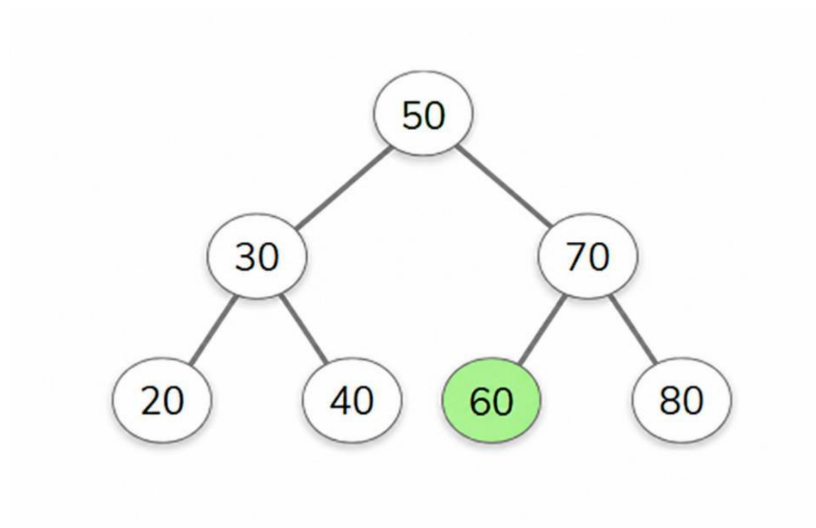
1. ConstructaBinarySearchTree(BST)byinsertingthefollowingkeys:50,30,70,20,40,60,80

   Perform:

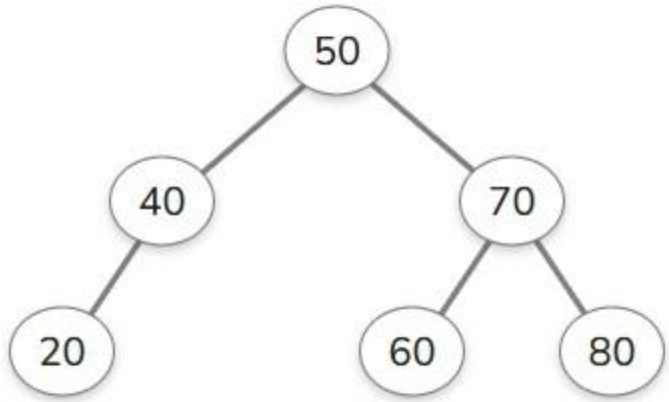
   a. **InorderTraversal**



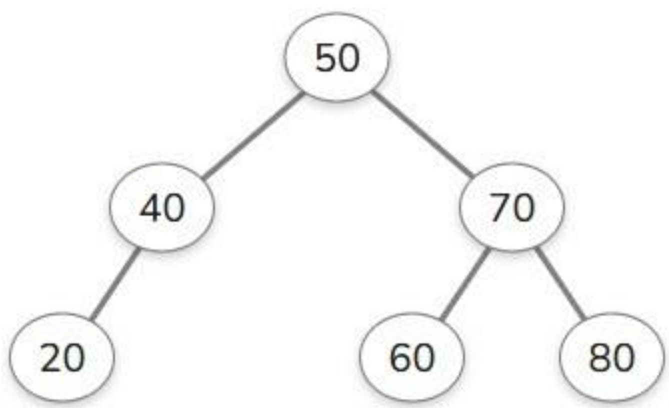
   b. **Searchfor60**



c. **Delete 30**

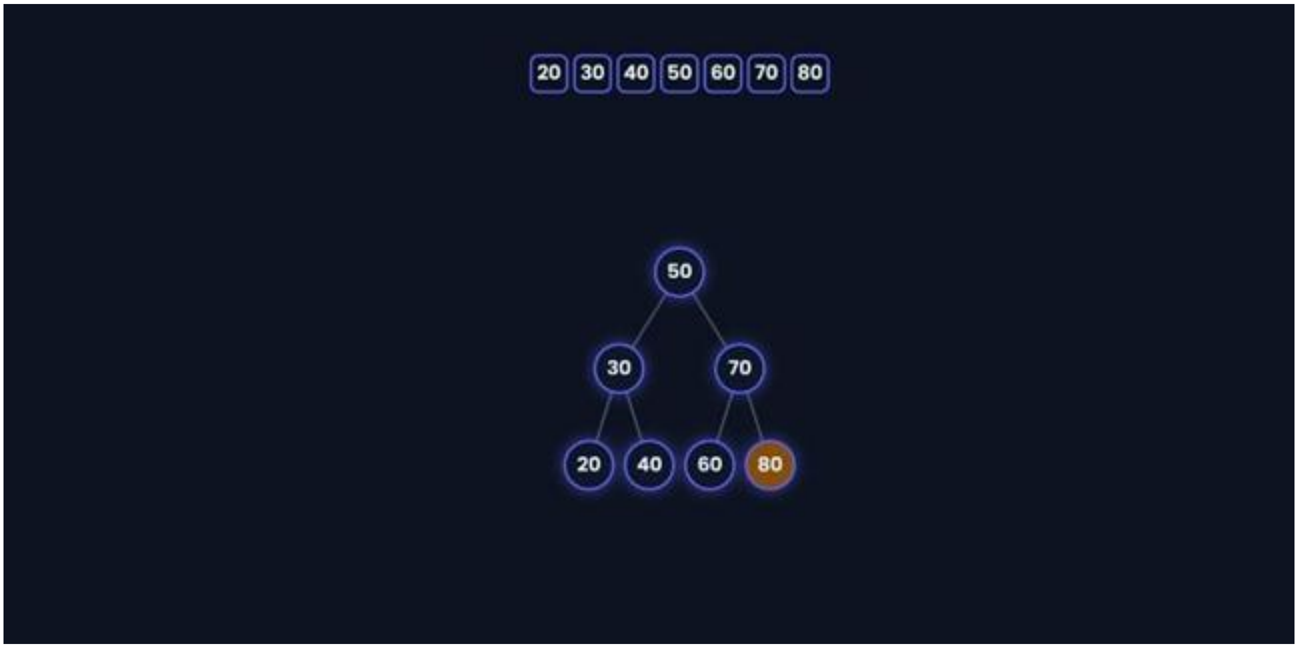


d. **DisplaythefinalBST.**

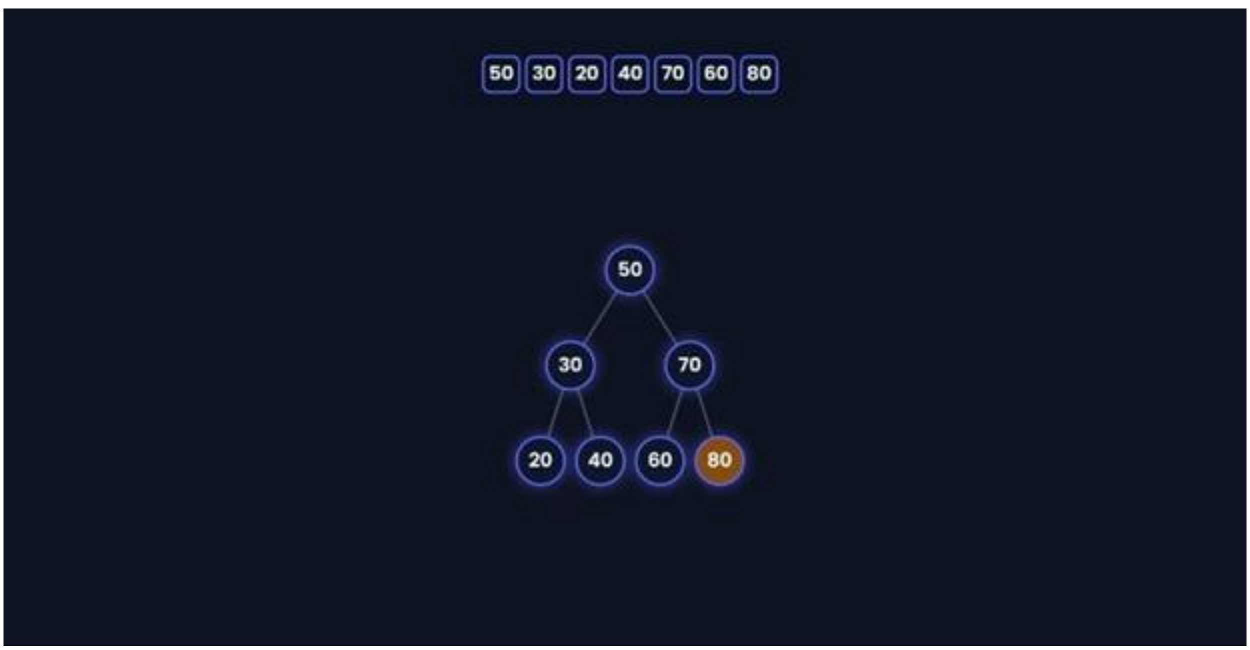


2. CompleteTreeTraversalsbyInserting50,30,70,20,40,60,80.Findallthetraversals.

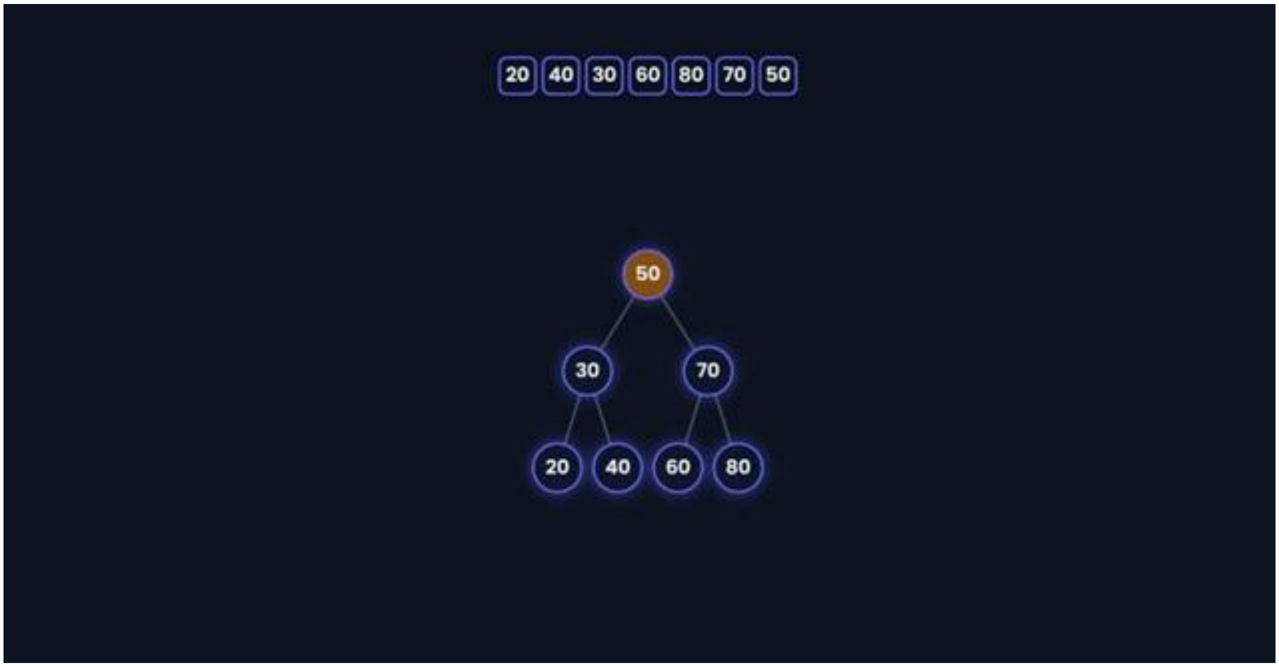
a. **Inorder**



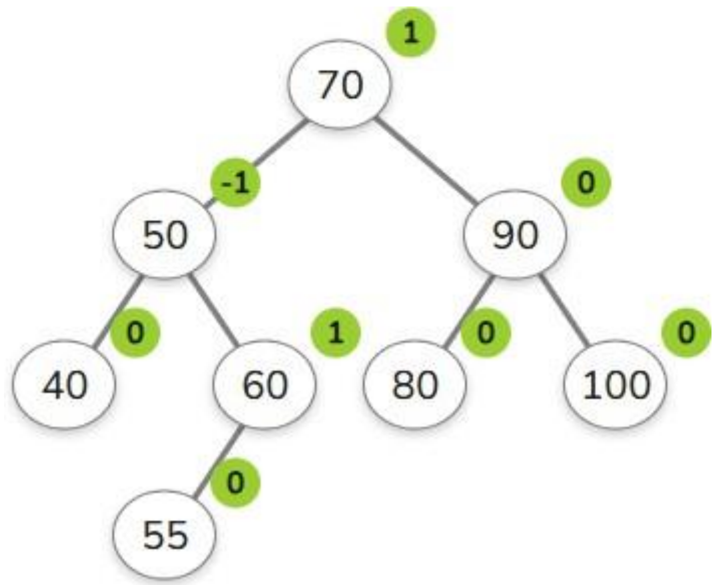
b. **Preorder**

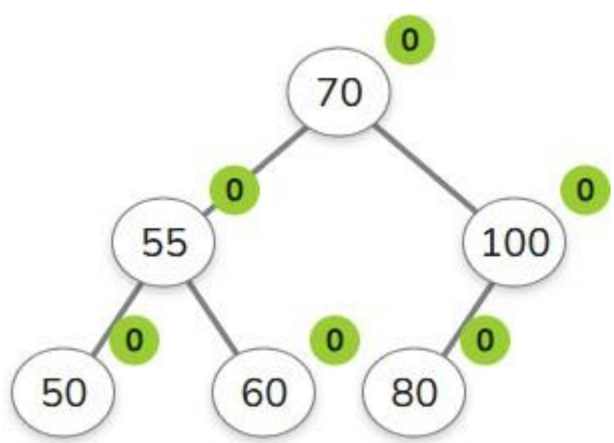


c. **Postorder**

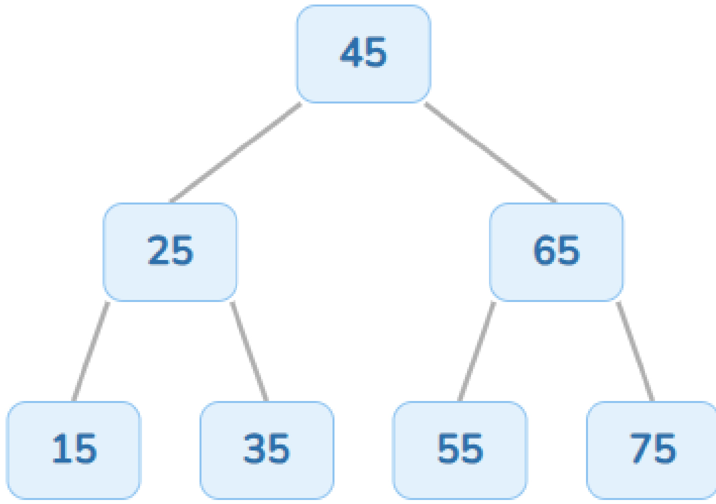


3. Insert70,50,90,40,60,80,100,55intoanAVLTree.Delete40and90.Analyzehowthe tree rebalances after each deletion.

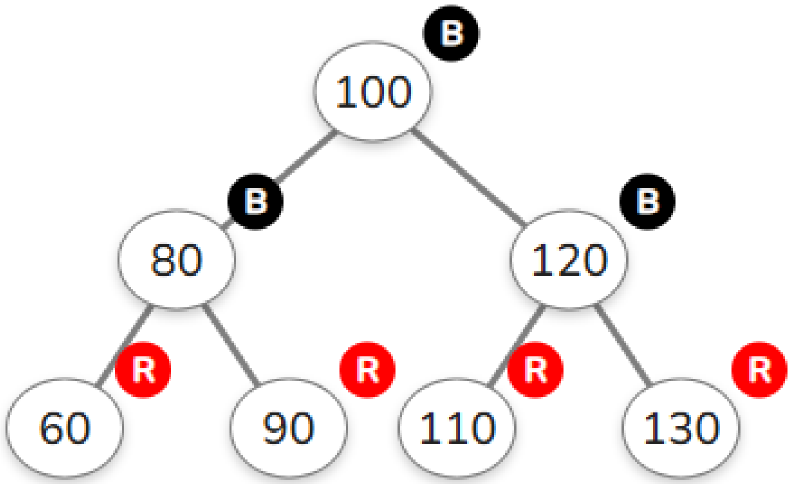




4. ConstructaB-Treeoforder3byinserting45,25,65,15,35,55,75



5. Insert100,80,120,60,90,110,130intoaRed-BlackTree.Analyzetheirheights.



Insertion	Tree Height
100	0
80	1
120	1
60	2
90	2
110	2
130	2